Han Bao

email | personal website

Github | Google Scholar

Chengdu, Sichuan - China

Education

- **Sichuan University** — *Sep 2021 - June 2026*
  Bachor of Cyber Science and Engineering — Chengdu, China
  - GPA: 3.56/4.00

Internship

- **University of Notre Dame [🌐]** — *Aug 2024 - Now*
  Visiting Student — South Bend, U.S.A.
  - Supervisor: Prof. Xiangliang Zhang
  - Research focus: Trustworthy Generative Model, Natural Language Processing

Publications

C=Conference, J=Journal, S=In Submission

[J.1] **CRISP: A Causal Relationships-Guided Deep Learning Framework for Advanced ICU Mortality Prediction**
Linna Wang, Xinyu Guo, **Han Bao**, Lihua Jiang, Li Zhao, Tao Zhu, Ziliang Feng
Accepted by *BMC Medical Informatics and Decision Making*.

[C.1] **LiD-FL: Towards List-Decodable Federated Learning**
Hong Liu, Liren Shan, **Han Bao**, Ronghui You, Yuhao Yi, Jiancheng Lv
Accepted by *AAAI Conference on Artificial Intelligence (AAAI 2025)*.

[S.1] **AutoDavis: Automatic Benchmarking of Large Vision-Language Models on Visual Question-Answering**
Han Bao*, Yue Huang*, Yanbo Wang, Jiayi Ye, Xiangqi Wang, Xiuying Chen, Mohamed Elhoseiny, Xiangliang Zhang
Preprint on arXiv. (*Equal Contribution)

[S.2] **On the Trustworthiness of Generative Foundation Models: Guideline, Assessment, and Perspective**
Yue Huang, Chujie Gao, Siyuan Wu, Haoran Wang, Xiangqi Wang, Yujun Zhou, Yanbo Wang, Jiayi Ye, Yuan Li, Jiawen Shi, Qihui Zhang, **Han Bao**, Zhaoyi Liu, Tianrui Guan, Dongping Chen, Ruoxi Chen, Kehan Guo, Kaijie Zhu, Taiwei Shi, Yiwei Li et al.
Preprint on arXiv. Project page: trustgen.github.io. ***Role: Leader of T2I Group.***

[S.3] ***PolicyLLM*: Towards Excellent Comprehension of Public Policy for Large Language Models**
Han Bao, Penghao Zhang, Yue Huang, Rui Su, Zhengqing Yuan, Yujun Zhou, Xiangqi Wang, Kehan Guo, Nitesh Chawla, Xiangliang Zhang
Manuscript in preparation.

[S.4] **Emergent Deceptive Behaviors in Reward-Optimizing LLMs**
Yujun Zhou*, **Han Bao***, Yue Huang, Kehan Guo, Zhenwen Liang, Pin-Yu Chen, Tian Gao, Werner Geyer, Nuno Moniz, Nitesh V Chawla, Xiangliang Zhang
Under Review. (*Equal Contribution)

Projects

- **Web-based Advanced Medical Image Processing and Analysis Platform** — *Oct. 2023 – Sep. 2024*
  Tools: Segment Anything Model (SAM), Glance, JavaScript, WebAssembly
  - Utilized the SAM (Segment Anything Model) for interactive and accurate medical image segmentation.
  - Replaced traditional OHIF viewer with the modern Glance viewer to improve image rendering speed and usability.
  - Built a web-based interface for clinicians to upload, segment, and review medical scans efficiently.
- **IntraAI: A Platform for Enhancing User Understanding of Large Language Models** — *June. 2025 – Sep. 2025*
  Tools: Large Language Model, Tag-based Personalization, Visualization
  - Built a platform that enables tag-based personalization and awareness-level probing to enhance user understanding of LLM outputs.
- **Uncertainty-Driven Meta-cognitive Learning Framework for LLMs** — *Sep. 2024 – Sep. 2025*
  Tools: PyTorch, Hugging Face Transformers, DPO, Sentence-BERT

- Proposed and designed AAH-Bench, the first benchmark to evaluate model behavior based on a 2x2 input/output uncertainty matrix.
- Developed an Uncertainty-Aware DPO (UA-DPO) algorithm that dynamically adjusts the training loss based on quantified uncertainty metrics.
- Designed a comprehensive evaluation suite including a Behavioral Alignment Score (BAS) to assess if a model exhibits the appropriate metacognitive strategy (e.g., clarification, refusal).

Honors and Awards

- **Provincial Third Prize – China Undergraduate Mathematical Contest in Modeling** *Nov. 2023*
  Chinese Society for Industrial and Applied Mathematics (CSIAM)
  - Demonstrated advanced problem-solving, mathematical modeling, and collaborative research skills.
- **Second-Class Scholarship** *Sep. 2023*
  School of Cyber Science and Engineering, Sichuan University
  - Awarded for academic excellence and overall performance within the top 10% of students.
  - Recognized for achievements in coursework, research potential, and campus engagement.

Skills

- **Programming & Development:** C, C++, Python, HTML, CSS, MySQL, Redis, Git, GitHub, Docker, Wireshark
- **Machine Learning & AI:** PyTorch, Transformers, LLaMA-Factory, LoRA fine-tuning, Reinforcement Learning (RL, VERL framework)
- **Research:** Literature Review, Paper Writing, Data Visualization
- **Languages:** IELTS 7.0 (English), Native Mandarin Chinese